Mitochondrial Genomic Signatures of Mammalian Longevity

Inspired by Farré et al. (2021), Mol. Biol. Evol.

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Short Abstract

Motivation
Comparative genomics can reveal genomic patterns associated with maximum lifespan. Farré et al. (2021) performed a large-scale nuclear genome analysis to identify convergent amino-acid substitutions (CAAS) linked to human lifespan.

Aim
In this project I adapt their conceptual framework, but focus exclusively on **mitochondrial DNA**. Using complete mitochondrial genomes from mammals, I search for amino-acid changes that distinguish long-lived from short-lived species.

Approach
I (1) compute body-mass–corrected Longevity Quotient (LQ) from AnAge data, (2) classify species into long-, short-, and intermediate-lived groups, (3) extract 13 mitochondrial protein-coding genes from GenBank genomes, (4) align genes at nucleotide and protein levels, and (5) detect and statistically evaluate CAAS.

Outcome
The pipeline produces a set of candidate mitochondrial amino-acid substitutions enriched in long-lived mammals, together with validation statistics and summary files for downstream biological interpretation.

Base Paper: Farré et al. 2021

Reference
Farré, X. et al. (2021). *Comparative Analysis of Mammal Genomes Unveils Key Genomic Variability for Human Life Span*. Molecular Biology and Evolution, 38(11):4948–4961.
`https://pmc.ncbi.nlm.nih.gov/articles/PMC8557403/`

Key ideas from the paper

- Large comparative study of mammalian **nuclear genomes** and lifespan.
- Definition of **Convergent Amino-Acid Substitutions (CAAS)** separating long-lived and short-lived species.
- Use of **Longevity Quotient (LQ)** to correct lifespan for body mass.
- Discovery and validation framework: discovery on extreme species, validation on intermediate species, plus permutation tests.

What I change
I keep the CAAS + LQ concept, but apply it entirely to **mitochondrial genomes** instead of whole nuclear genomes.

Project Aim and Research Questions

Overall Aim

- Identify mitochondrial amino-acid patterns that are associated with extended lifespan in mammals.

Specific questions

- Do long-lived mammals share specific amino acids at mitochondrial protein positions that differ from short-lived mammals?
- Are these patterns statistically enriched compared with random species groupings (permutation test)?
- Which mitochondrial genes (ND, COX, ATP, CYTB) are most enriched for such CAAS?

Scope

- Only complete mitochondrial genomes.
- 13 canonical mitochondrial protein-coding genes.
- Focus on mammals with available lifespan and body mass data (AnAge).

Data Overview

Phenotype data (AnAge)

- Maximum lifespan (years).
- Body mass (g).
- Taxonomy: Class, Order, Family, Genus, Species.

Genomic data (NCBI GenBank)

- Complete mitochondrial genomes downloaded via NCBI Entrez.
- One GenBank file per species (`.gb`).

Genes analysed (mitochondrial)

- ATP synthase: **ATP6, ATP8**
- Cytochrome c oxidase: **COX1, COX2, COX3**
- Cytochrome b: **CYTB**
- NADH dehydrogenase: **ND1, ND2, ND3, ND4, ND4L, ND5, ND6**

Method: Longevity Quotient (LQ)

Goal

Correct lifespan for body mass to compare "how long" species live relative to their size.

Steps (implemented in `anage_LQ.py`)

- Filter to mammals: `Class = Mammalia`.
- Compute body mass in kg:

$$\text{body\_mass\_kg} = \frac{\text{Body mass (g)}}{1000}$$

- Expected maximum lifespan:

$$\text{expected\_mls} = 4.88 \times \text{body\_mass\_kg}^{0.19}$$

- Longevity Quotient:

$$\text{LQ} = \frac{\text{max\_lifespan\_years}}{\text{expected\_mls}}$$

- Define species ID: `Genus_Species` and store `longevity_quotient`.

Grouping

Species are ranked by LQ:

- Top tail: **long-lived group**.
- Bottom tail: **short-lived group**.
- Middle region: **validation group**.

Method: From GenBank to Alignments

Downloading genomes (`download_genbank.py`)

- Read target species from `data/LQ/fetch_targets.txt`.
- Query NCBI `nuccore` for complete mitochondrial genomes (RefSeq preferred).
- Save GenBank files to `data/genbank_files/`.

Gene extraction (`extract_genes_genbank.py`)

- Parse CDS annotations in each GenBank file.
- Normalize gene names (e.g. COI/COXI $\rightarrow$ COX1, COB $\rightarrow$ CYTB, NAD1 $\rightarrow$ ND1).
- Extract nucleotide sequences for each of the 13 genes.
- Translate using vertebrate mitochondrial genetic code (table 2), remove terminal stops.
- Output:
    - Nucleotides: `data/extracted_genes/nucleotides/GENE/species.fasta`
    - Proteins: `data/extracted_genes/proteins/GENE/species.fasta`

Method: Alignment and CAAS Discovery

Alignment (`align_genes.py`)

- For each gene, combine per-species FASTA into a multi-FASTA.
- Align with MAFFT (default) or MUSCLE.
- Produce:
  - `data/alignments/proteins/GENE_aligned.fasta`
  - `data/alignments/nucleotides/GENE_aligned.fasta`
- Compute simple quality metrics (number of sequences, alignment length, gap percentage).

CAAS discovery (`caas_discovery_from_lists.py`)

- Use predefined long-lived and short-lived lists from `data/LQ`.
- At each alignment column, collect residues for all long- and short-lived species (no gaps, all present).
- Test three scenarios (following Farré et al.):
  - S1: long fixed (A), short fixed (B), $A \neq B$.
  - S2: long fixed (A), short variable, but never A.
  - S3: short fixed (B), long variable, but never B.

Method: Validation and Permutation Testing

Validation with intermediate species

- For each CAAS position:
  - In validation species, separate those with the "long-lived" amino acid vs the "short-lived" amino acid (or pattern).
  - Use Longevity Quotient (from `lq_results.csv`) for each validation species.
  - Apply **one-sided Mann–Whitney U test** to see if the long-like configuration has higher LQ.
- Mark CAAS as validated when P-value is below a chosen threshold (e.g. 0.2 as implemented).

Permutation test

- Randomly shuffle species between long-lived and short-lived groups (keeping group sizes).
- Recompute number of S1 + S2 CAAS for each permutation.
- Compare the observed count to the null distribution to get an empirical P-value.
- If observed CAAS $\gg$ null expectation, this supports a real longevity signal.

Results: Overview of CAAS Counts

Example summary (protein-level)

Gene	Scenario 1	Scenario 2	Scenario 3
ATP6	1	4	0
COX1	2	1	1
ND5	5	3	5
⋮			

Notes

- Fill this table with the real counts from `caas_discovered_protein.csv` or your JSON summary.
- Highlight genes with the highest CAAS density (e.g. ND genes if that is the case).

Interpretation (high level)

- Genes with many CAAS may be more involved in lifespan-related mitochondrial functions.

Results: CAAS Validation and Significance

Validation summary

- Total discovered CAAS (protein): **51**.
- CAAS with sufficient validation data: **51**.
- Validated CAAS (P $<$ threshold): **6**.
- Validation rate: **11%**.

Permutation test

- Observed S1+S2 CAAS: **34**.
- Mean null S1+S2: **1.1**.
- Empirical P-value: **0.004**.

Visualisation idea

- Add a barplot or histogram comparing observed vs null CAAS counts:

(Insert your histogram / barplot figure here)

Results: Example CAAS Positions

Example table of validated CAAS

Gene	Position	Scenario	Long AA / Short AA	P-value
ND2	14	S2	[FI] / [L]	
COX3	175	S3	[FL] / [I]	
⋮				

How to fill this slide

- Select a few biologically interesting positions from `caas_validated_protein.csv` or `caas_validated_nucleotide.csv`.
- Highlight if they occur in functional domains or conserved regions (if you check that later).

Thank You!